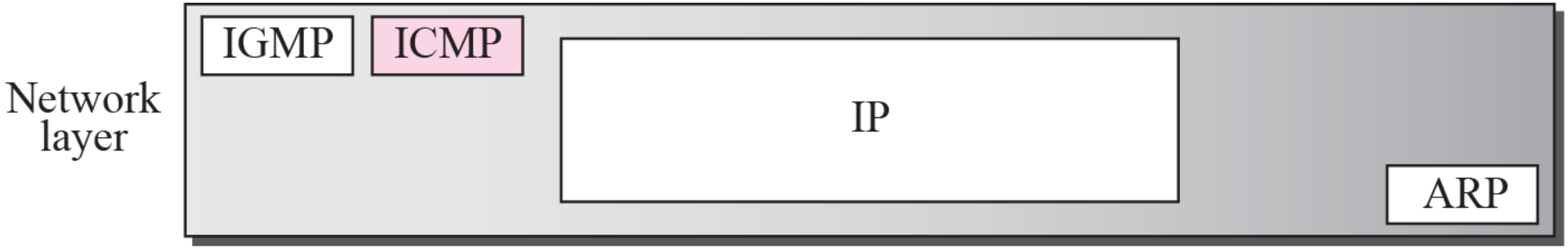
EETS 7302 TCP/IP Network Administration

Session 7
Internet Control Message Protocol
ICMPv4

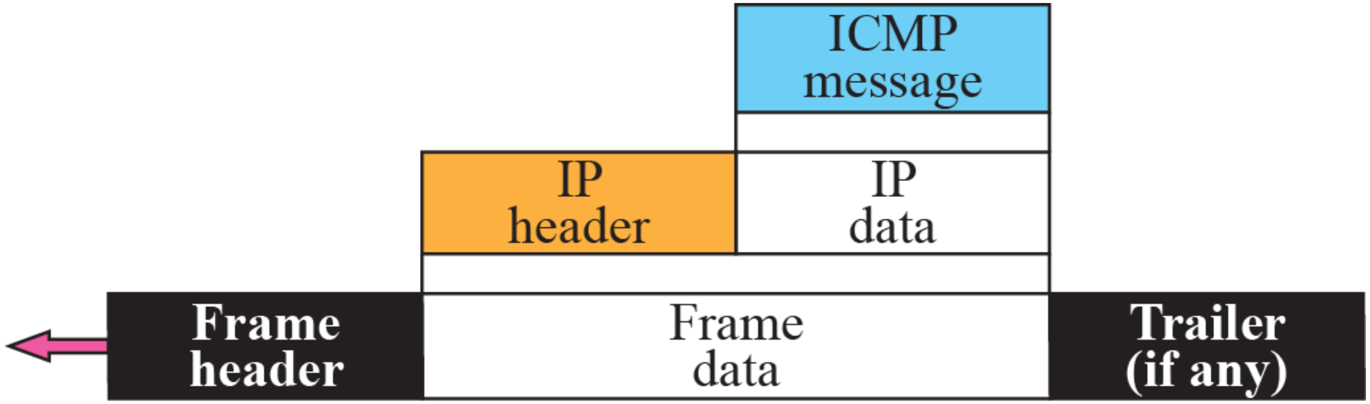
INTRODUCTION

- **IP protocol has no error-reporting or error correcting mechanism.**
  - **What happens if something goes wrong?**
  - **What happens if a router must discard a datagram because it cannot find a router to the final destination, or because the time-to-live field has a zero value?**

Position of ICMP in the network layer



ICMP encapsulation



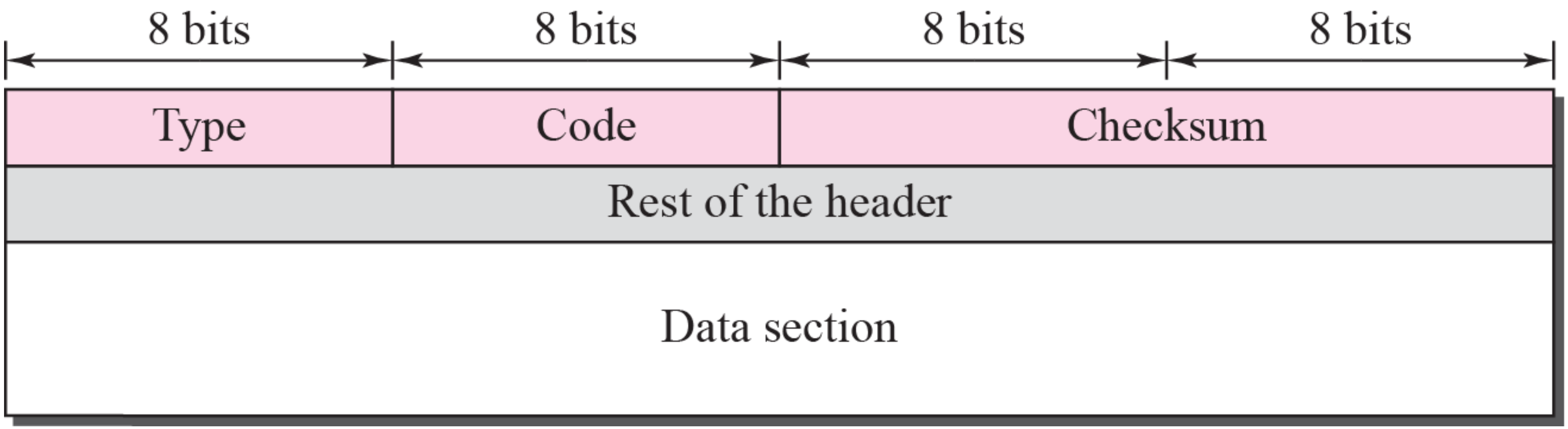
ICMP MESSAGES

- ICMP messages are divided into two broad categories:
  - error-reporting messages: report problems when it processes an IP packet
  - query messages: occur in pairs, help a host or a network manager get specific information from a router or another host. Also, hosts can discover and learn about routers on their network and routers can help a node redirect its messages.

Table 9.1 *ICMP messages*

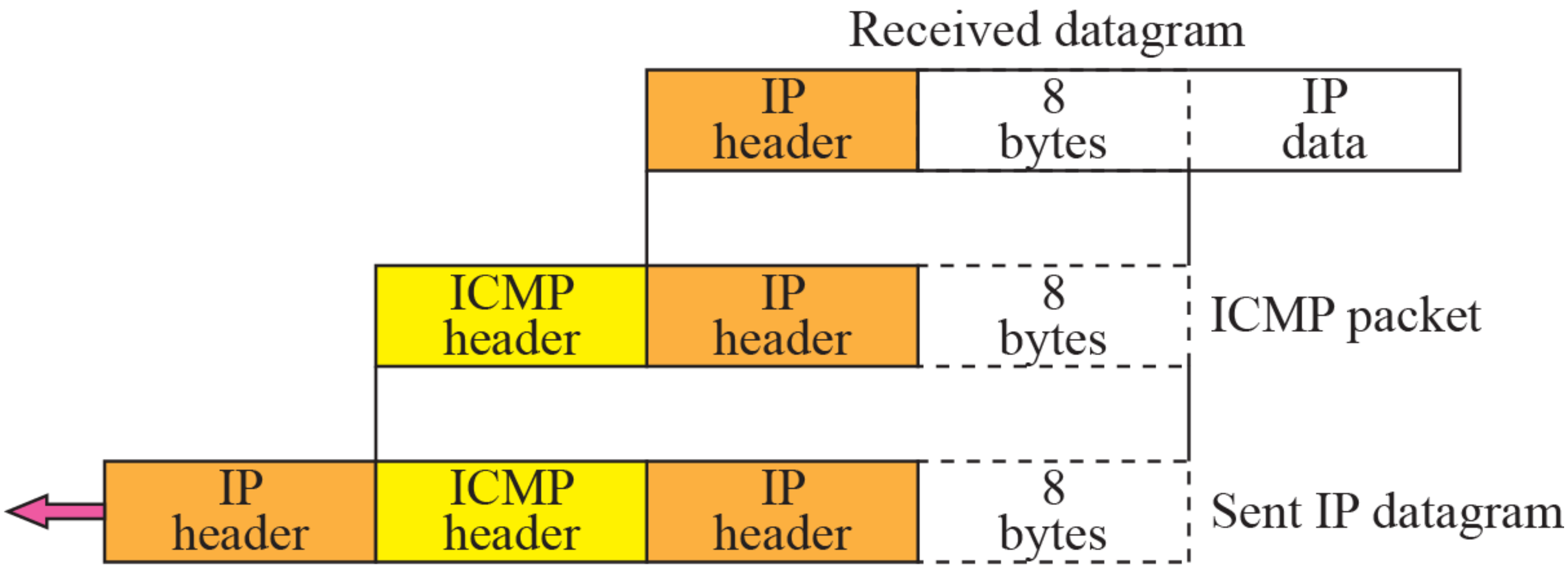
Category	*Type*	*Message*
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply

General format of ICMP messages



ICMP always reports error messages to the original source.

Contents of data field for the error message



Destination-unreachable format

Type: 3	Code: 0 to 15	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Source-quench format

Type: 4	Code: 0	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

A source-quench message informs the source that a datagram has been discarded due to congestion in a router or the destination host.

The source must slow down the sending of datagrams until the congestion is relieved.

Time-exceeded message format

Type: 11	Code: 0 or 1	Checksum
Unused (All 0s)		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

Whenever a router decrements time-to-live value to zero, it discards the datagram and sends a time-exceeded message to the original source.

When destination does not receive all fragments in a set time, it discards the received fragments and sends a time-exceeded message to the original source.

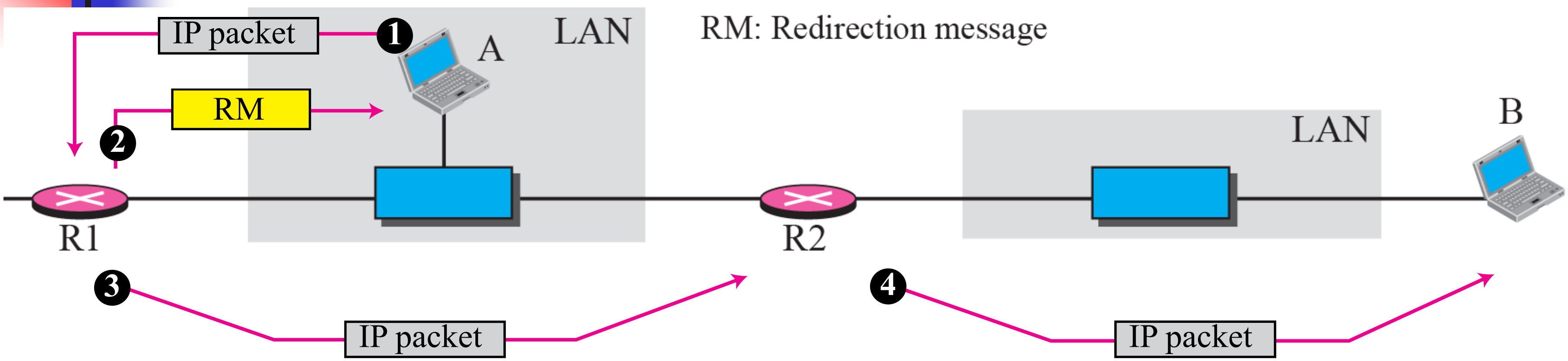
In a time-exceeded message, code 0 is used only by routers to show that the value of the time-to-live field is zero.

Code 1 is used only by the destination host to show that not all of the fragments have arrived within a set time.

Parameter-problem message format

Type: 12	Code: 0 or 1	Checksum
Pointer	Unused (All 0s)	
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

A parameter-problem message can be created by a router or the destination host.



A host usually starts with a small routing table that is gradually augmented and updated.

One of the tools to accomplish this is the redirection message.

Type: 5	Code: 0 to 3	Checksum
IP address of the target router		
Part of the received IP datagram including IP header plus the first 8 bytes of datagram data		

A redirection message is sent from a router to a host on the same local network.

Echo-request and echo-reply message

- Type 8: Echo request
- Type 0: Echo reply

Type: 8 or 0	Code: 0	Checksum
Identifier		Sequence number
Optional data Sent by the request message; repeated by the reply message		

An echo-request message can be sent by a host or router. An echo-reply message is sent by the host or router that receives an echo-request message.

Echo-request and echo-reply messages can be used by network managers to check the operation of the IP protocol.

Echo-request and echo-reply messages can test the reachability of a host. This is usually done by invoking the ping command.

Timestamp-request and timestamp-reply message format

- Type 13: request
- Type 14: reply

Type: 13 or 14	Code: 0	Checksum	
Identifier		Sequence number	
Original timestamp			
Receive timestamp			
Transmit timestamp			

Timestamp-request and timestamp-reply messages can be used to calculate the round-trip time between a source and a destination machine even if their clocks are not synchronized.

The timestamp-request and timestamp-reply messages can be used to synchronize two clocks in two machines if the exact one-way time duration is known.

DEBUGGING TOOLS

- several tools that can be used in the Internet for debugging.
- Help find if a host or router is alive and running.
- trace the route of a packet.
- two tools that use ICMP for debugging:
    - ping
    - traceroute.

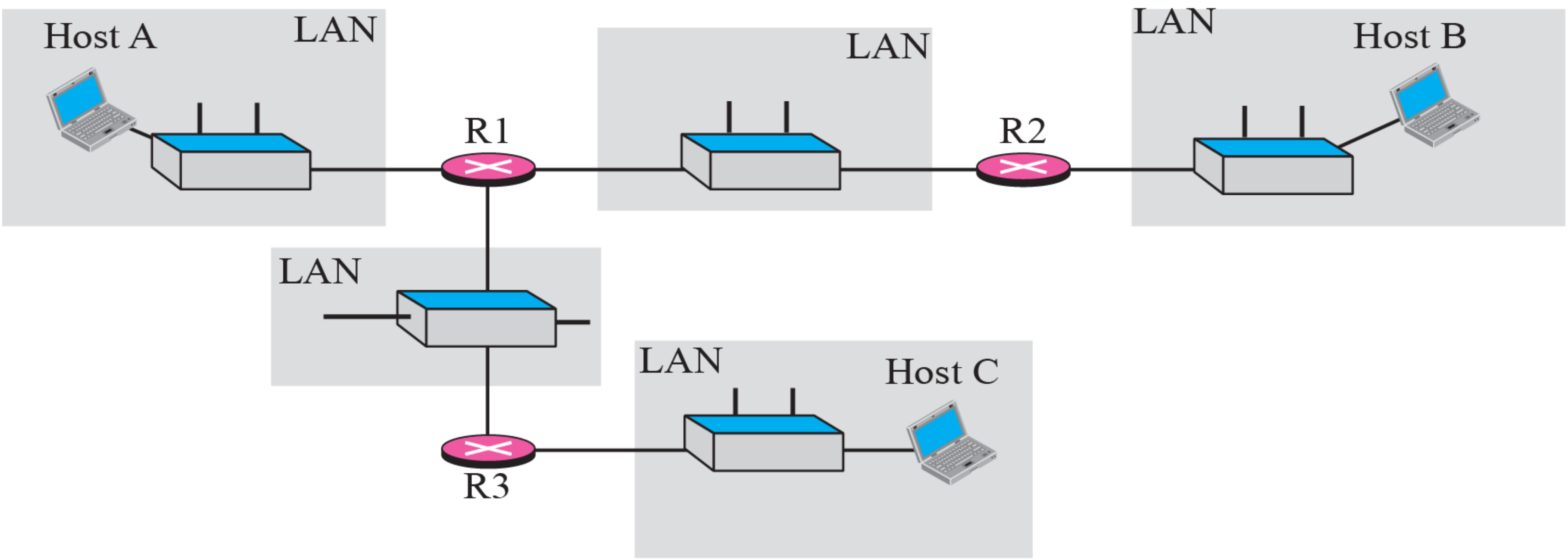
Example

use the ping program to test the server fhda.edu.

```
$ ping fhda.edu
PING fhda.edu (153.18.8.1)    56 (84)  bytes of data.
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=0    ttl=62    time=1.91 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=1    ttl=62    time=2.04 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=2    ttl=62    time=1.90 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=3    ttl=62    time=1.97 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=4    ttl=62    time=1.93 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=5    ttl=62    time=2.00 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=6    ttl=62    time=1.94 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=7    ttl=62    time=1.94 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=8    ttl=62    time=1.97 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=9    ttl=62    time=1.89 ms
64 bytes from tiptoe.fhda.edu (153.18.8.1): icmp_seq=10   ttl=62    time=1.98 ms

--- fhda.edu ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10103 ms
rtt min/avg/max = 1.899/1.955/2.041 ms
```

The traceroute program operation



Example

use the traceroute program to find the route from the computer voyager.deanza.edu to the server fhda.edu.

```
$ traceroute fhda.edu
traceroute to fhda.edu      (153.18.8.1), 30 hops max, 38 byte packets
1 Dcore.fhda.edu            (153.18.31.25)    0.995 ms    0.899 ms    0.878 ms
2 Dbackup.fhda.edu          (153.18.251.4)    1.039 ms    1.064 ms    1.083 ms
3  tiptoe.fhda.edu           (153.18.8.1)     1.797 ms    1.642 ms    1.757 ms
```

Example

In this example, we trace a longer route, the route to xerox.com. The following is a partial listing.

```
$ traceroute xerox.com
traceroute to xerox.com (13.1.64.93), 30 hops max, 38 byte packets
1  Dcore.fhda.edu    (153.18.31.254)    0.622 ms   0.891 ms   0.875 ms
2  Ddmz.fhda.edu     (153.18.251.40)    2.132 ms   2.266 ms   2.094 ms
3  Cinic.fhda.edu    (153.18.253.126)   2.110 ms   2.145 ms   1.763 ms
4  cenic.net         (137.164.32.140)   3.069 ms   2.875 ms   2.930 ms
5  cenic.net         (137.164.22.31)    4.205 ms   4.870 ms   4.197 ms
6  cenic.net         (137.164.22.167)   4.250 ms   4.159 ms   4.078 ms
7  cogentco.com      (38.112.6.225)     5.062 ms   4.825 ms   5.020 ms
8  cogentco.com      (66.28.4.69)       6.070 ms   6.207 ms   5.653 ms
9  cogentco.com      (66.28.4.94)       6.070 ms   5.928 ms   5.499 ms
```

Example

An interesting point is that a host can send a traceroute packet to itself. This can be done by specifying the host as the destination. The packet goes to the loopback address.

```
$ traceroute voyager.deanza.edu
traceroute to voyager.deanza.edu   (127.0.0.1), 30 hops max, 38 byte packets
1  voyager        (127.0.0.1)        0.178 ms        0.086 ms        0.055 ms
```

Example

When traceroute does not receive a response within 5 seconds, it prints an asterisk to signify a problem (not the case in this example), and then tries the next hop.

```
$ traceroute mhhe.com
traceroute to mhhe.com (198.45.24.104), 30 hops max, 38 byte packets
1    Dcore.fhda.edu        (153.18.31.254)      1.025 ms    0.892 ms    0.880 ms
2    Ddmz.fhda.edu         (153.18.251.40)      2.141 ms    2.159 ms    2.103 ms
3    Cinic.fhda.edu        (153.18.253.126)     2.159 ms    2.050 ms    1.992 ms
4    cenic.net             (137.164.32.140)     3.220 ms    2.929 ms    2.943 ms
5    cenic.net             (137.164.22.59)      3.217 ms    2.998 ms    2.755 ms
6    SanJose1.net          (209.247.159.109)   10.653 ms   10.639 ms   10.618 ms
7    SanJose2.net          (64.159.2.1)        10.804 ms   10.798 ms   10.634 ms
8    Denver1.Level3.net    (64.159.1.114)      43.404 ms   43.367 ms   43.414 ms
9    Denver2.Level3.net    (4.68.112.162)      43.533 ms   43.290 ms   43.347 ms
10   unknown               (64.156.40.134)     55.509 ms   55.462 ms   55.647 ms
11   mcleodusa1.net        (64.198.100.2)      60.961 ms   55.681 ms   55.461 ms
12   mcleodusa2.net        (64.198.101.202)    55.692 ms   55.617 ms   55.505 ms
13   mcleodusa3.net        (64.198.101.142)    56.059 ms   55.623 ms   56.333 ms
14   mcleodusa4.net        (209.253.101.178)  297.199 ms  192.790 ms  250.594 ms
15   eppg.com              (198.45.24.246)     71.213 ms   70.536 ms   70.663 ms
16   ...                   ...                 ...         ...         ...
```

References

TCP/IP Protocol Suite (4th Edition),

by Behrouz A. Forouzan – Used the text's slide deck

End of session